

AI Resume Screening System

Name

Kohinoor Soni

Minor project 1

AI Resume Screening System using Machine Learning and NLP

Introduction:

Recruitment is one of the most important processes in any organization. Companies often receive hundreds or thousands of resumes for a single job opening. Manually reviewing all these resumes is time consuming and inefficient.

The AI Resume screening System is designed to automate the resume evaluation process using Machine Learning and Natural Language Processing (NLP). The system analyzes resumes, extracts useful information, predicts the candidate's job category, identifies technical skills, calculates resume quality, and evaluates how well the resume matches a given job description.

The project helps recruiters shortlist suitable candidates faster and also helps job seekers understand how well their resume fits for specific job roles.

Problem statement:

Recruiters spend significantly time manually reviewing resumes. Traditional screening methods are slow, inconsistent, and prone to human bias.

The objective of this project is to develop an intelligent system that can:

- Automatically classify resume into job categories.
- Extract important skills from resumes.
- Calculate a resume quality score.
- Match resumes with job description using ATS (Application tracking system) principles.
- Display prediction through an interactive web application.

The system should reduce recruiter workload and improve candidate screening efficiency.

Objectives:

The main objective of the project are:

1. Automate resume classification.
2. Extract skills from resumes.
3. Analyze candidate profile.
4. Calculate resume quality scores.
5. Perform ATS matching with job description.
6. Display top predicted job categories.
7. Build an easy-to-use web interface.

Technology stack:

Programming Language

- Python 3.11

Library Used

- Pandas
- Numpy

Machine Learning

- Scikit-learn

NLP

- NLTK
- Regular expression (regex).

Model Storage

- Joblib

PDF Processing

- PyPDF2

Web Application

- Streamlit

Dataset Information:

Dataset Name:

UpdatedResumeDataSet.csv

Dataset Size:

- 962 resumes
- 25 job categories

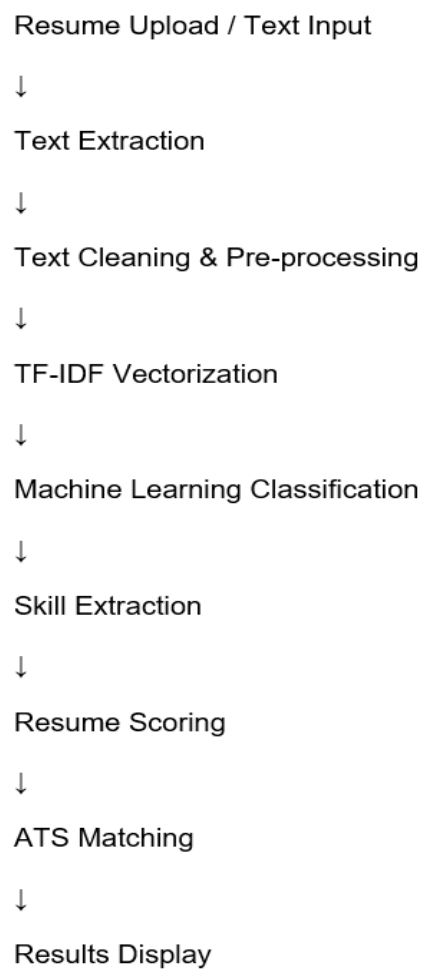
Examples of categories:

- Data Science
- Python Developer
- Java Developer
- DevOps Engineer
- Testing
- Blockchain
- HR
- Web Designing
- Mechanical Engineer
- Electrical Engineering

The dataset contains:

- Resume Text
- Job Category

System Architecture



Data Pre-processing

Before training the model, the resumes are cleaned using NLP techniques.

The following preprocessing steps are performed:

Lowercase Conversion

All text is converted to lowercase.

URL Removal

Website links are removed.

Special Character Removal

Symbol and unwanted character are removed.

Whitespace Removal

Extra Spaces are removed

Features Engineering

TF-IDF (Term Frequency-Inverse Document Frequency) is used to convert text into Numerical vectors.

Configuration:

- Max_feature = 5000
- Ngram_range = (1,2)

Output Shape:

(962, 5000)

This means every resume is represented using 5000 important features.

Machine Learning Model

Algorithm Used:

Logistic Regression

Parameters:

- Max_iter = 5000

Reason for selection:

- Fast training

- Good performance on text classification
- Works well with TF-IDF features
- Easy to interpret

Model Training

The dataset was divided into training and testing sets using an 80:20 ratio. The Logistic Regression model was trained on the processed TF-IDF features extracted from the resume dataset.

Training Configuration:

- Training Data: 80%
- Testing Data: 20%
- Random State: 42
- Maximum Iterations: 5000

Model Performance

The trained model achieved an accuracy of 99.48% on the test dataset.

Performance Metrics:

- Accuracy: 99.48%
- Precision: 99%
- Recall: 100%
- F1-Score: 99%

These results indicate that the model performs effectively in classifying resumes into their respective job categories.

Features Implemented

The system includes the following features:

- Resume Classification
- PDF Resume Upload
- Skill Extraction
- Resume Score Calculation
- ATS Match Score
- Top 3 Category Predictions
- Streamlit Web Interface

Streamlit Web Application:

A user-friendly web application was developed using Streamlit.

The application allows users to:

- Upload PDF resumes
- Paste resume text
- Predict job categories
- View confidence scores
- Extract technical skills
- Calculate resume scores
- Perform ATS matching
- View top predicted categories

Results

The system successfully classified resumes into appropriate job categories and extracted relevant skills from uploaded resumes.

Sample Output:

- Predicted Category: Data Science
- Confidence Score: 26.54%
- Resume Score: 100/100
- ATS Match Score: Based on Job Description Match

AI Resume Screening System

Upload a PDF Resume or Paste Resume Text

Upload Resume PDF

PDF uploaded successfully!

View Extracted Text

Resume Text

Job Description (For ATS Matching)

Predict Category

Predicted Category: Data Science

Confidence Score: 26.54%

Resume Score

Resume Score: 100.00/100

Detected Skills

- ✓ c++
- ✓ git
- ✓ java
- ✓ javascript
- ✓ machine learning
- ✓ mysql
- ✓ python
- ✓ sql

Top 3 Predictions

- ✓ Data Science — 26.54%
- ✓ Blockchain — 7.16%
- ✓ Python Developer — 5.81%

The application provides quick and meaningful insights regarding candidate profiles.

Conclusion

The AI Resume Screening System automates resume analysis using Machine Learning and NLP techniques. The system can classify resumes, extract skills, calculate resume scores, and perform ATS matching through an interactive web application.

Using TF-IDF feature extraction and logistic Regression, the model achieved an accuracy of 99.48%, demonstrating strong performance in resume classification tasks. The project provides an efficient solution for modern recruitment and candidate screening processes.

References:

1. Scikit-Learn Documentation
2. NLTK Documentation
3. Streamlit Documentation
4. Pypdf2 Documentation
5. Kaggle Resume Dataset
6. Machine learning and NLP research papers